

CI/CD Pipeline for Brain Tasks Application using AWS Services

Aim

To build and deploy a containerized Brain Tasks application using AWS CodeBuild, Amazon ECR, Amazon EKS, Kubernetes, Docker, and CloudWatch.

Tools Used

- GitHub
 - Docker
 - AWS CodeBuild
 - Amazon ECR
 - Amazon EKS
 - Kubernetes
 - CloudWatch
 - kubectl
-

Architecture Diagram

GitHub Repository

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AWS CodeBuild

↓

Docker Image Build

↓

Amazon ECR

↓

Amazon EKS

↓

Kubernetes Deployment



LoadBalancer Service



Brain Tasks Application

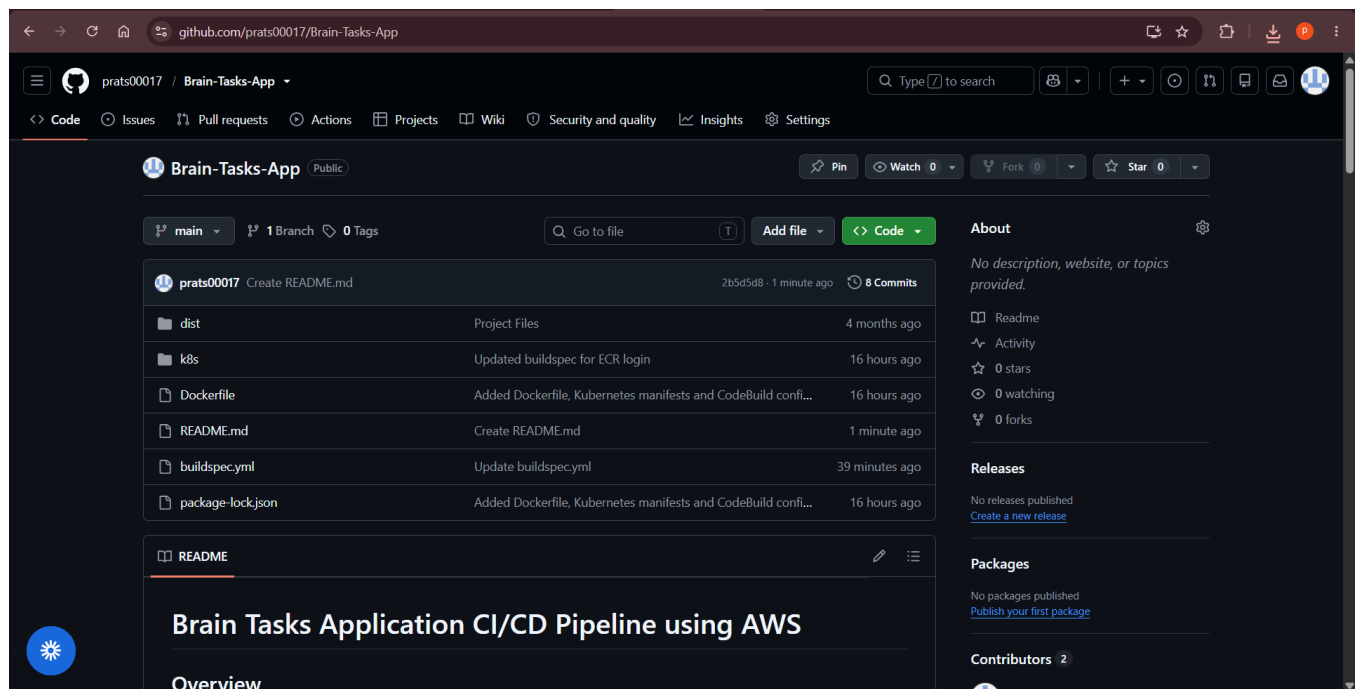


CloudWatch Logs

Step 1: Source Code Repository

The Brain Tasks application source code was maintained in GitHub.

Screenshot:



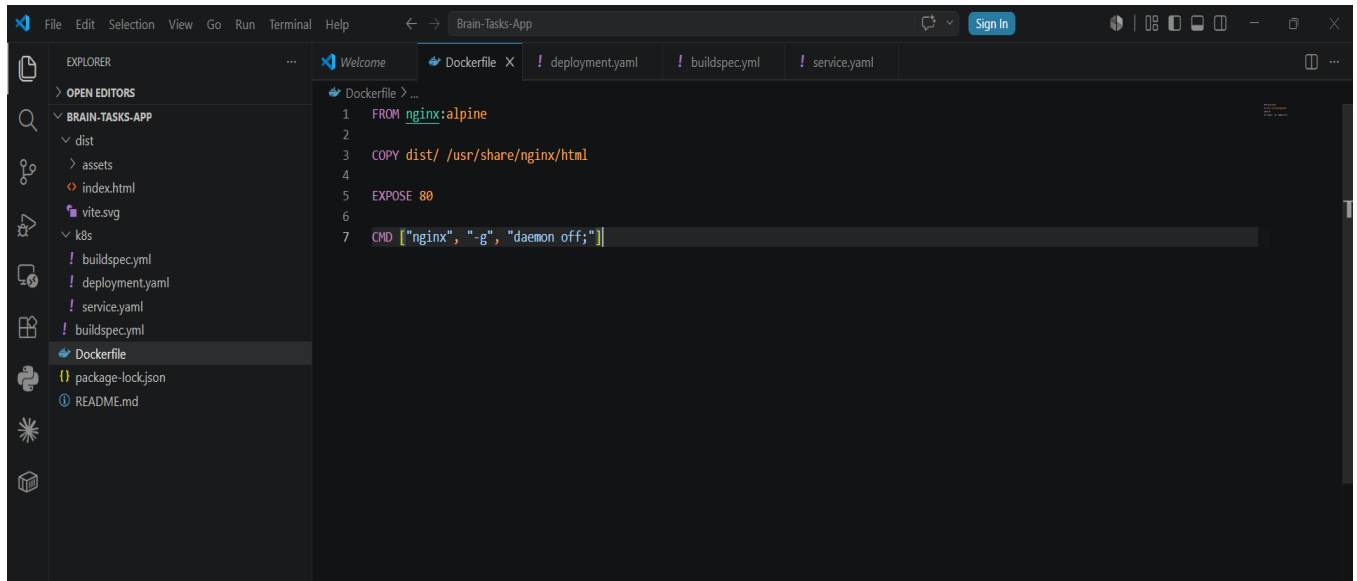
Step 2: Dockerization

A Dockerfile was created to containerize the application.

Command:

```
docker build -t brain-tasks-app .
```

Screenshot:



Step 3: AWS CodeBuild

Created project:

brain-tasks-codebuild

Configured buildspec.yml and integrated GitHub.

Build completed successfully.

Screenshot:

The screenshot displays the AWS CodeBuild console interface. The top navigation bar includes the AWS logo, a search bar, and the user's account information (prats0017). The left sidebar shows the 'Developer Tools' menu with 'CodeBuild' selected. The main content area is titled 'brain-tasks-codebuild' and features several action buttons: 'Actions', 'Create trigger', 'Edit', 'Clone', 'Debug build', 'Start build with overrides', and 'Start build'. Below these buttons is a 'Configuration' section with the following details:

- Source provider: GitHub
- Primary repository: prats0017/Brain-Tasks-App
- Artifacts upload location: -
- Service role: arn:aws:iam::360357042190:role/service-role/codebuild-brain-tasks-codebuild-service-role
- Public builds: Disabled

Below the configuration section is a 'Build history' tab, which is currently selected. It shows a table of build runs with the following columns: Build run, Status, Build number, Source version, Submitter, Duration, and Completed. The table contains one entry:

Build run	Status	Build number	Source version	Submitter	Duration	Completed
brain-tasks-codebuild-9454709-9-5a5d-4b2a-bfc3-69c9c9136de2	Succeeded	5	main	root	30 seconds	34 minutes ago

The bottom of the console shows the footer with copyright information (© 2026, Amazon Web Services, Inc. or its affiliates) and links to Privacy, Terms, and Cookie preferences.

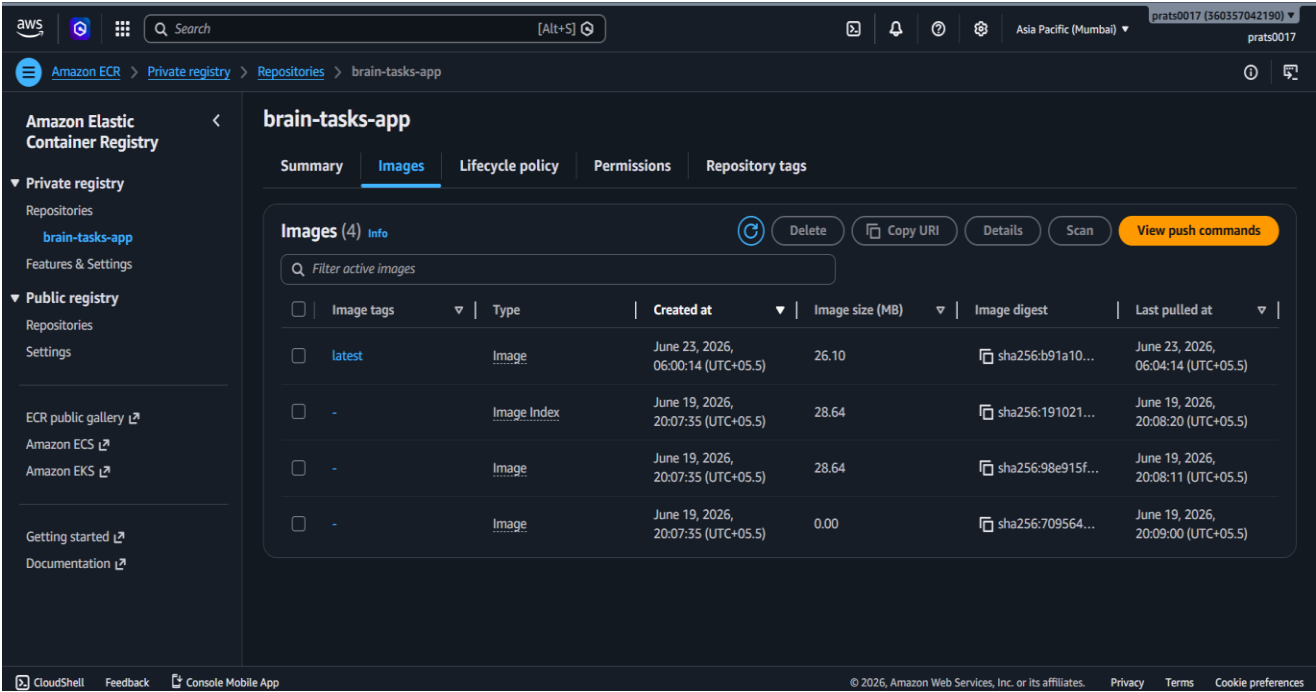
Step 4: Amazon ECR

Repository:

brain-tasks-app

Docker image was pushed successfully.

Screenshot:



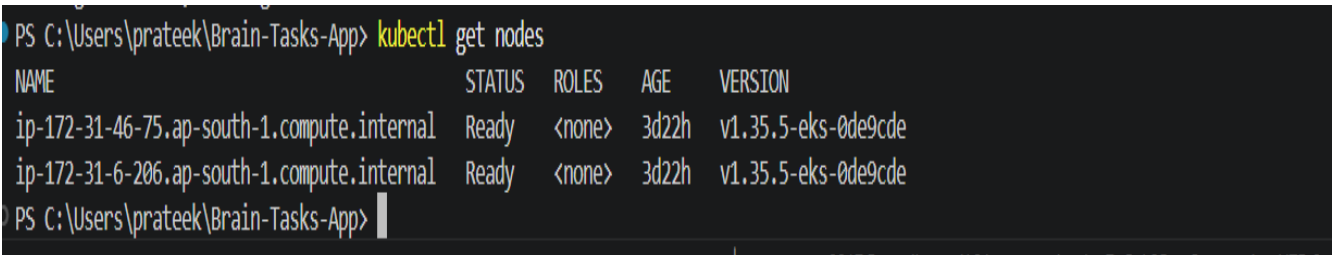
Step 5: Amazon EKS

Cluster nodes verified.

Command:

```
kubectl get nodes
```

Screenshot:



Step 6: Deployment

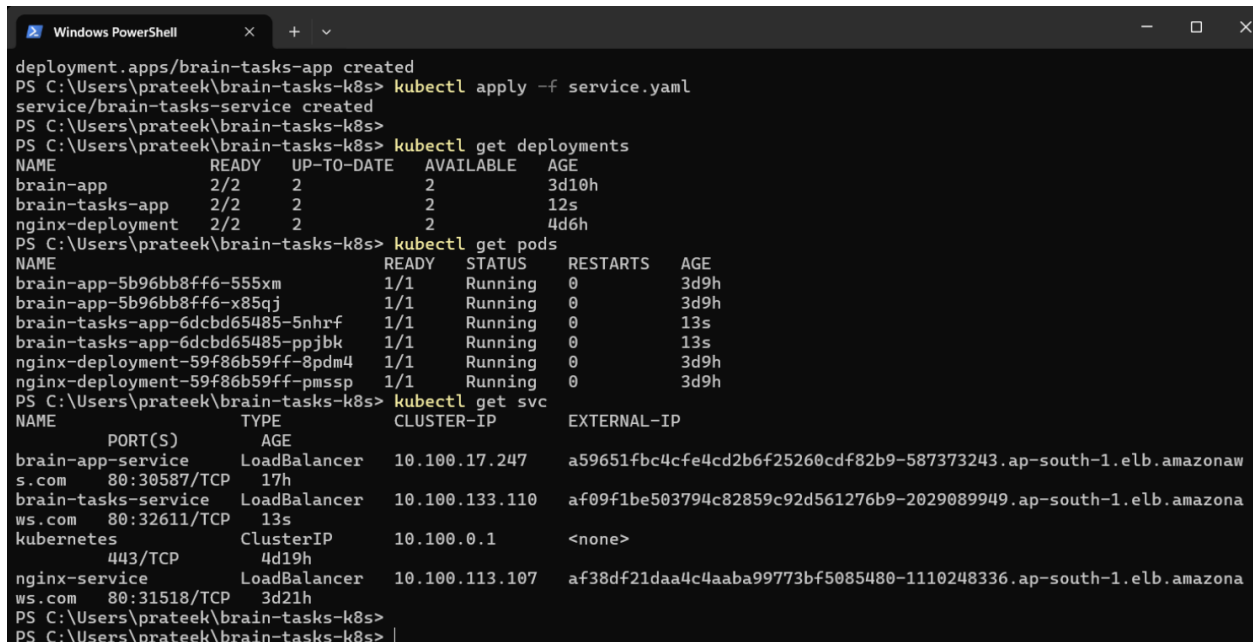
Applied Kubernetes manifests.

Commands:

```
kubectl apply -f deployment.yaml
```

```
kubectl apply -f service.yaml
```

Screenshot:



```
deployment.apps/brain-tasks-app created
PS C:\Users\prateek\brain-tasks-k8s> kubectl apply -f service.yaml
service/brain-tasks-service created
PS C:\Users\prateek\brain-tasks-k8s>
PS C:\Users\prateek\brain-tasks-k8s> kubectl get deployments
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
brain-app            2/2     2             2           3d10h
brain-tasks-app      2/2     2             2           12s
nginx-deployment     2/2     2             2           4d6h
PS C:\Users\prateek\brain-tasks-k8s> kubectl get pods
NAME                READY   STATUS    RESTARTS   AGE
brain-app-5b96bb8ff6-555xm    1/1     Running   0           3d9h
brain-app-5b96bb8ff6-x85qj    1/1     Running   0           3d9h
brain-tasks-app-6dcdb65485-5nhrf  1/1     Running   0           13s
brain-tasks-app-6dcdb65485-ppjbk  1/1     Running   0           13s
nginx-deployment-59f86b59ff-8pdm4  1/1     Running   0           3d9h
nginx-deployment-59f86b59ff-pmssp  1/1     Running   0           3d9h
PS C:\Users\prateek\brain-tasks-k8s> kubectl get svc
NAME                TYPE        CLUSTER-IP      EXTERNAL-IP
--                --
brain-app-service   LoadBalancer  10.100.17.247   a59651fbc4cfe4cd2b6f25260cdf82b9-587373243.ap-south-1.elb.amazonaws.com
brain-tasks-service LoadBalancer  10.100.133.110 af09f1be503794c82859c92d561276b9-2029089949.ap-south-1.elb.amazonaws.com
kubernetes          ClusterIP     10.100.0.1      <none>
nginx-service       LoadBalancer  10.100.113.107 af38df21daa4c4aaba99773bf5085480-1110248336.ap-south-1.elb.amazonaws.com
PS C:\Users\prateek\brain-tasks-k8s>
PS C:\Users\prateek\brain-tasks-k8s> |
```

Step 7: Pods Verification

Command:

```
kubectl get pods
```

Screenshot:

As above

Step 8: Services Verification

Command:

```
kubectl get svc
```

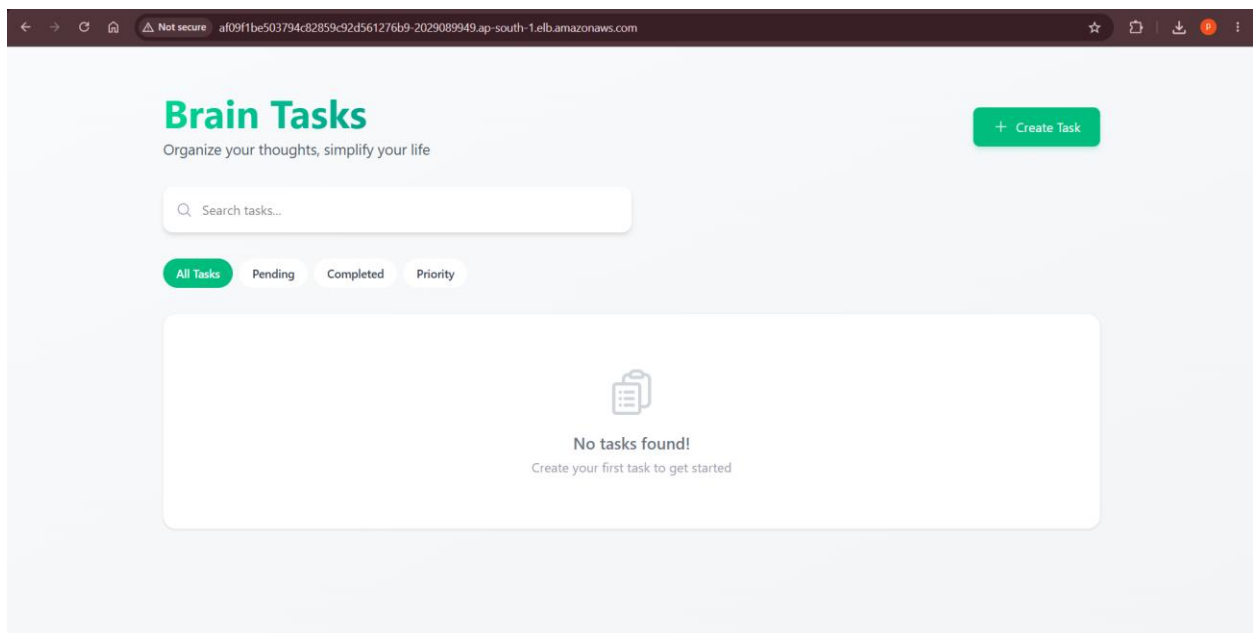
Screenshot:

As above

Step 9: Application Output

Application was accessed through the external LoadBalancer URL.

Screenshot:



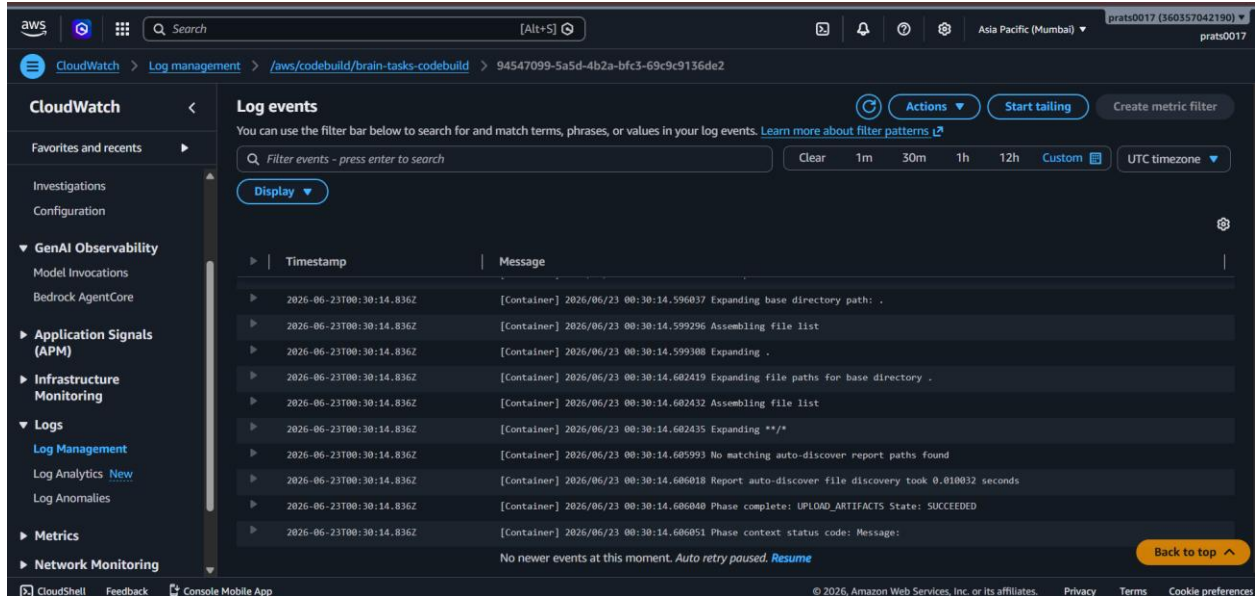
Step 10: CloudWatch Monitoring

CloudWatch log group:

```
/aws/codebuild/brain-tasks-codebuild
```

Build logs were monitored successfully.

Screenshot:



Commands Used

```
kubectl get nodes
kubectl get deployments
kubectl get pods
kubectl get svc
kubectl apply -f deployment.yaml
kubectl apply -f service.yaml
```

Result

Successfully implemented and deployed the Brain Tasks application using a complete CI/CD pipeline with GitHub, AWS CodeBuild, Amazon ECR, Amazon EKS, Kubernetes, and CloudWatch.

Conclusion

The project demonstrates an end-to-end DevOps workflow using AWS cloud services. Continuous integration and deployment were achieved successfully, and the application

was deployed and exposed externally using Kubernetes LoadBalancer with monitoring support through CloudWatch.